



Customer Shopping Behavior Analysis

Transactional data from **3,900 purchases** across product categories — uncovering spending patterns, segments, and preferences.

Dataset at a Glance

3,900

Rows

18

Columns

37

Missing Values

Review Rating

\$59.76

Avg Purchase

Key features: demographics, purchase details, shopping behavior, and subscription status.

Data Preparation in Python

→ Clean & Standardize

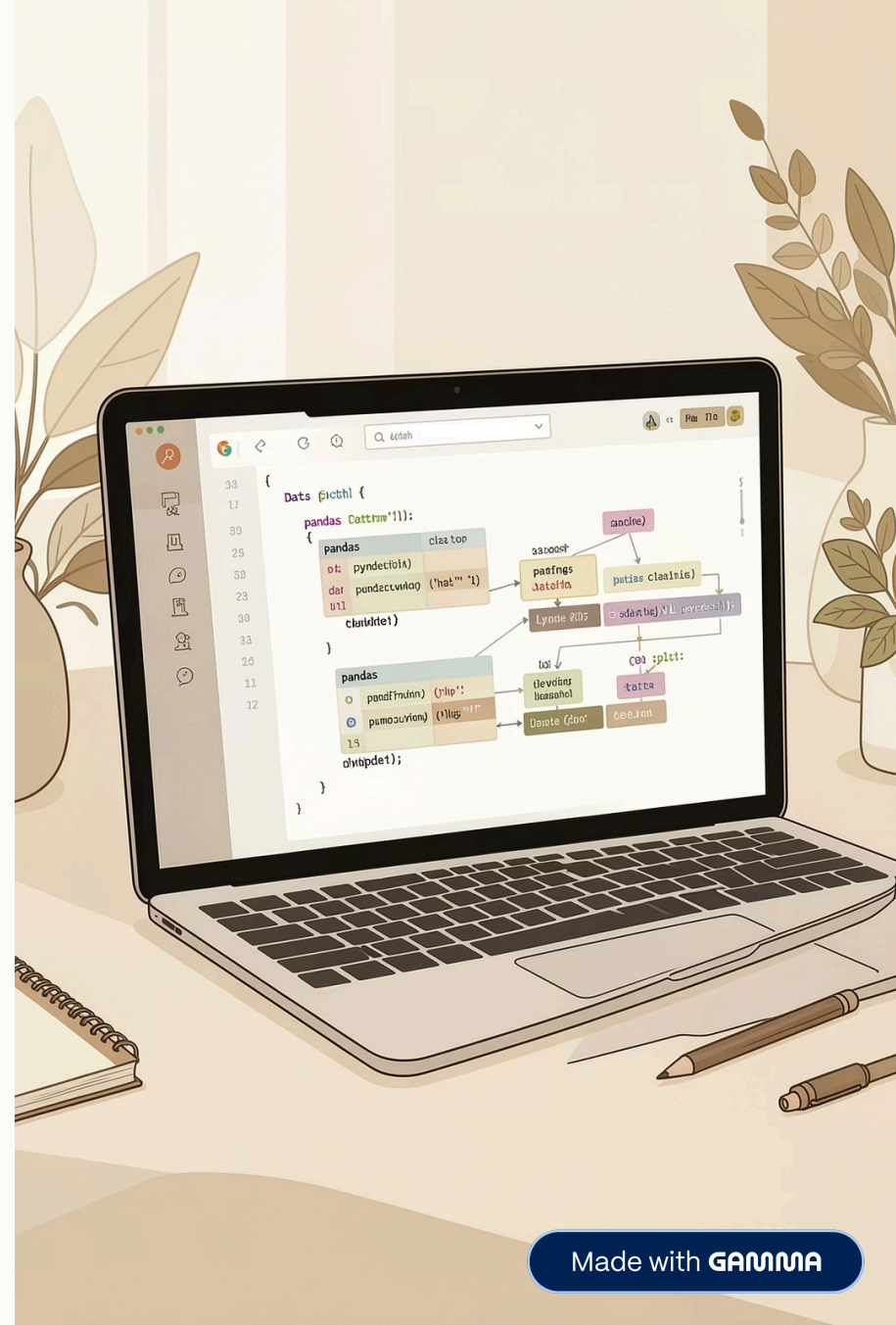
Snake_case columns, median imputation for missing ratings

→ Feature Engineering

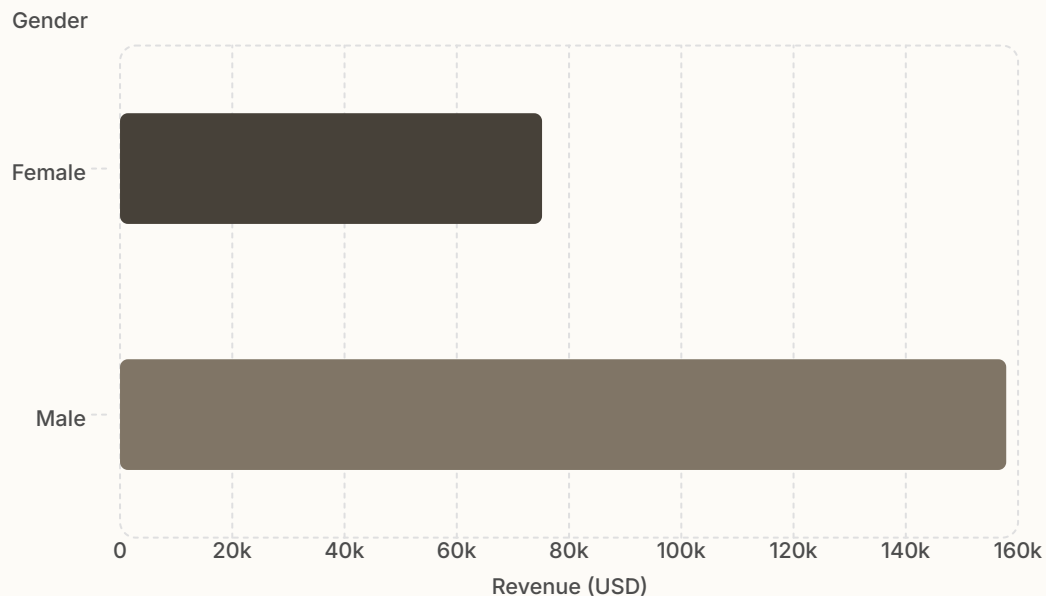
Age groups, purchase frequency days

→ Database Integration

Loaded cleaned DataFrame into PostgreSQL



Revenue by Gender



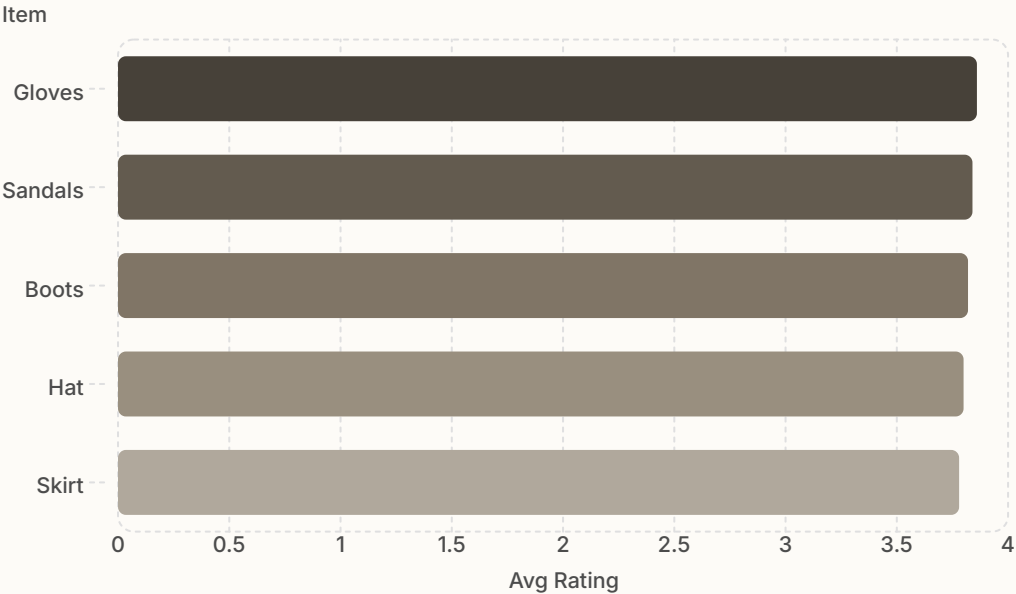
Key Insight

Male customers generated **2.1× more revenue** than female customers across all categories.

i Male: \$157,890 · Female: \$75,191

Top Products & Shipping Insights

Top 5 Products by Rating

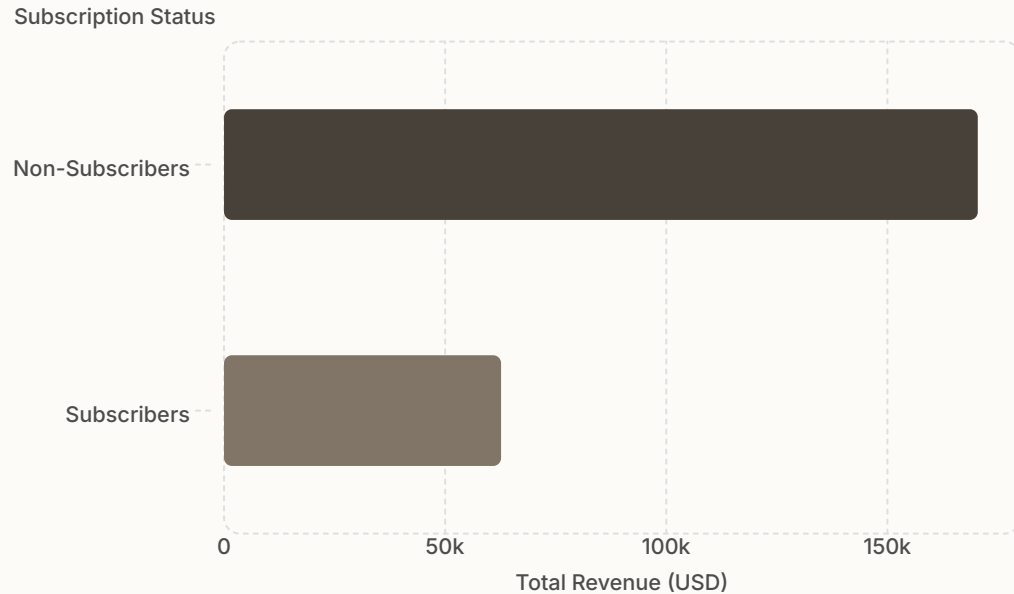


Shipping Type Comparison



Express shipping customers spend slightly more on average.

Subscribers vs. Non-Subscribers



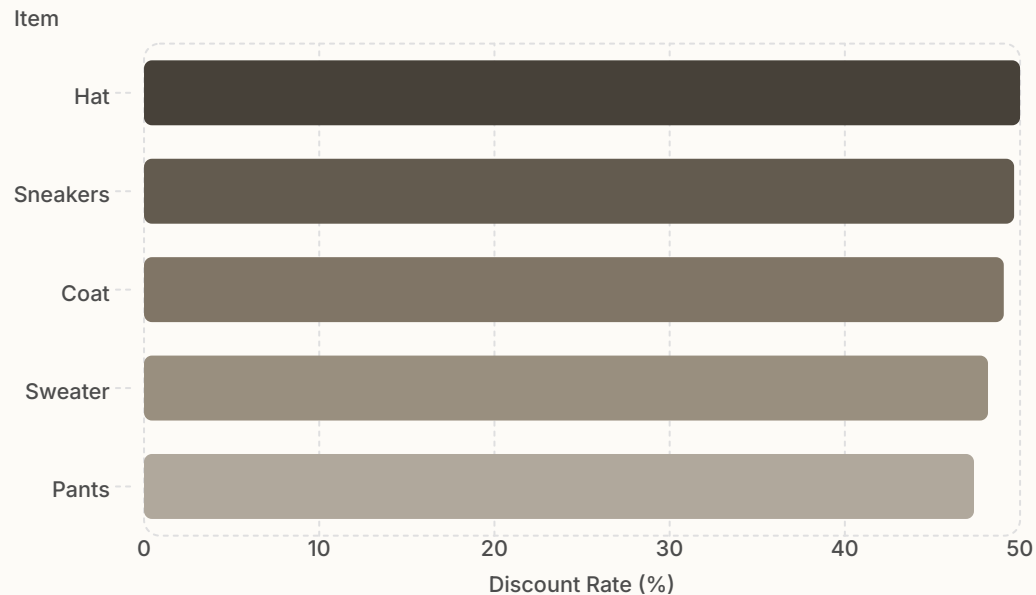
Key Findings

- Non-subscribers: **2,847 customers**, avg spend \$59.87
- Subscribers: **1,053 customers**, avg spend \$59.49
- Non-subscribers drive **73%** of total revenue

⚠ Only 27% of customers are subscribed — major growth opportunity.

Discount-Dependent Products & Customer Segments

Top 5 Discount-Dependent Products



Customer Segmentation

Loyal

3,116 customers

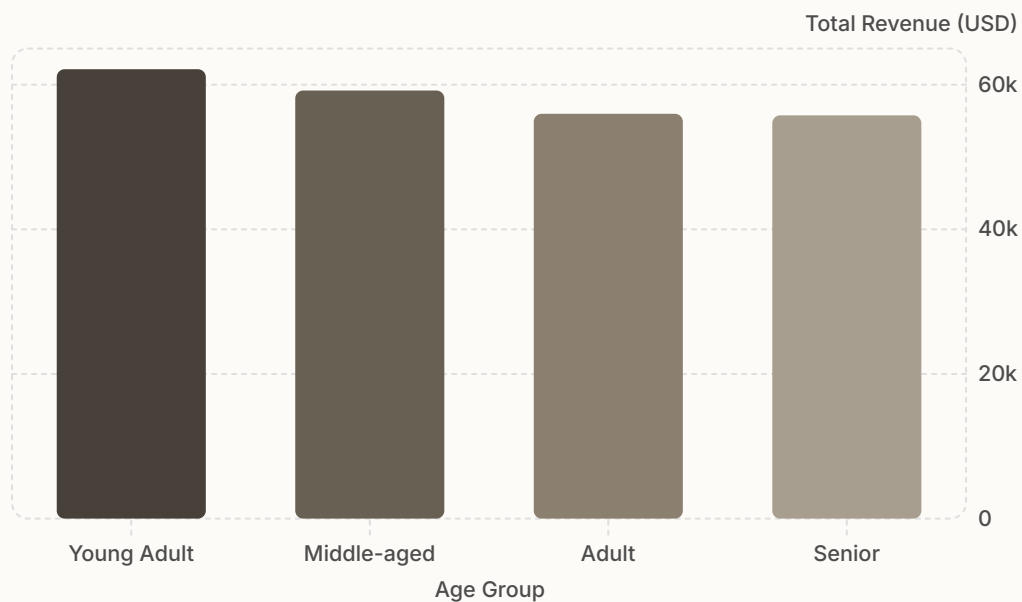
Returning

701 customers

New

83 customers

Revenue by Age Group



Age Group Insights

Revenue is relatively balanced across age groups, with **Young Adults** leading at \$62,143.

- All four groups contribute within ~\$6K of each other
- Young Adults represent the highest-revenue segment

Power BI Dashboard

3.9K Customers

Total customer base

\$59.76 Avg Purchase

Average spend per order

3.75 Avg Rating

Average review score

Filters

Gender, Category, Subscription, Shipping



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers into Loyal segment



Review Discount Policy

Balance sales boosts with margin control



Targeted Marketing

Focus on high-revenue age groups & express users

